Data Visualisation Assignment

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2026-04-12

```
library(dplyr)
library(lubridate)
library(tidyr)
library(ggplot2)
library(osmdata)
library(sf)
library(viridis)
library(patchwork)
library(corrplot)
library(tinytex)
library(knitr)
library(ggforce)
library(ggrepel)
```

#Does Colchester Become a more Dangerous Place in the Summer? ##Introduction

Within this Data Visualisation the relationship between crime and the weather within Colchester in 2025 will be explored. This is to determine whether on a baseline level if more crimes occur within the hotter months of the year or whether weather has no impact upon how dangerous Colchester was in 2025. This will be explored through correlation plots, smoothed scatter graphs and boxplots with sina plots overlayed. With this combination of graphs and a solid understanding of the dataset we will conclude whether Colchester is more dangerous during the summer or not.

##Data Cleaning and Preparation

Within this section data will be manipulated in which to become more usable and aid the depth of analysis that can be performed within the visualisation, To begin data was loaded in and given appropriate names.

```
weather <- read.csv("temp25.csv")
crime <- read.csv("crime25.csv")
str(weather$Date)
```

```
##  chr [1:365] "2025-12-31" "2025-12-30" "2025-12-29" "2025-12-28" ...
```

```
str(crime$date)
```

```
##  chr [1:5956] "2025-01" "2025-01" "2025-01" "2025-01" "2025-01" "2025-01" ...
```

As shown above the dates are within an incorrect format having been stored as characters. Therefore we will need to change them into actual dates. This was done with help of the lubridate package. Due to the nature of the crime data already being in months an arbitrary date was created for each month to add the day variable. This date was created as the 1st of each month.

```
crime$date <- ym(crime$date)
weather$Date <- as.Date(weather$Date)
str(crime$date)
```

```
##  Date[1:5956], format: "2025-01-01" "2025-01-01" "2025-01-01" "2025-01-01" "2025-01-01" ...
```

```
str(weather$Date)
```

```
##  Date[1:365], format: "2025-12-31" "2025-12-30" "2025-12-29" "2025-12-28" "2025-12-27" ...
```

Here shows that the dates are now correctly formatted which will help with future analysis and interpretation.

```
str(weather$WindkmhDir)
```

```
##  chr [1:365] "NNW" "NNE" "NE" "NE" "NE" "ENE" "ENE" "ENE" "E" "E" "SE" "SW" ...
```

The direction of the wind is given in character form for which doing a visualisation would be very difficult. Therefore each direction along with its mapped wind speed was transformed into x and y vectors of wind. This was conducted by mapping each compass direction onto its degree equivalent. After that was produced it was then applied to the wind direction within the data and only numerical values were kept. Due to Rstudios cos and sin functions dealing with radians these degree measurements were then needed to be converted into radians. Once this was completed using vector formulae these radians were then combined with the wind speed to determine the velocity of the wind respective to x and y. This was then added back to the data.

```
dir_map <- c("N" = 0, "NNE" = 22.5,"NE" = 45,"ENE" = 67.5,"E" = 90,"ESE" = 112.5,"SE" = 135,"SSE" = "157

weather$wind_deg <- as.numeric(dir_map[weather$WindkmhDir])
weather$wind_rad <- weather$wind_deg * pi/180

xwind <- weather$WindkmhInt * cos(weather$wind_rad)
ywind <- weather$WindkmhInt * sin(weather$wind_rad)
weather$xwind <- xwind
weather$ywind <- ywind
```

Due to the differing natures of the data with weather being in days and crime data being in months an equilibrium was required to compare th data and determine the relationships. This equilibrium was decided as monthly data. This enables detailed analysis while not making up fraudulent data, that would be required in which to perform any less. To achieve this monthly data the weather data needs to be converted. Due to the varying nature of this data with some being cumulative and others being measurements 2 separate methods were used. On the measurements such as temperature, clouds and humidity the mean was applied. However to data of cumulative totals, such as precipitation and sun duration, summation was applied to better reflect their patterns. Due to a very high number of missing data, snow depth and atmospheric pressure were not included within the monthly data.

```
sum(is.na(weather$SnowDepcm))
```

```
## [1] 349
```

```r
sum(is.na(weather$PreselevHp))
```

```
## [1] 365
```

```r
weather$month <- floor_date(weather$Date, "month")
weather_monthly_means <- weather %>%
  group_by(month) %>%
  summarise(across(c(TemperatureCAvg,TemperatureCMax,TemperatureCMin,TdAvgC,HrAvg,lowClOct,TotClOct,Vis
```

```r
weather_monthly_sum <- weather %>%
  group_by(month) %>%
  summarise(across(c(Precmm,SunD1h),\(x) sum(x, na.rm = TRUE)))
```

Now both monthly weather data variables are combined to form one weather monthly dataset.

```r
weather_monthly <- left_join(weather_monthly_means,weather_monthly_sum,by = "month")
```

Within the crime dataset the variable category enables us to see what type each crime committed is. However to perform a good visualisation there are too many variables that are similar. In order to use the category data each category was put into one of 6 types as opposed to the 14 categories. This maintains each crimes category to a sufficient level of detail for analysis while enabling visualisation to be far easier.

```r
unique(crime$category)
```

```
##  [1] "anti-social-behaviour" "bicycle-theft"         "burglary"
##  [4] "criminal-damage-arson" "drugs"                 "other-theft"
##  [7] "possession-of-weapons" "public-order"          "robbery"
## [10] "shoplifting"           "theft-from-the-person" "vehicle-crime"
## [13] "violent-crime"         "other-crime"
```

```r
crime <- crime %>%
  mutate(Type = case_when(
    grepl("theft|shoplift|bicycle|vehicle|burglary",category,ignore.case = TRUE) ~ "Theft",
    grepl("violent|robbery|weapon",category,ignore.case = TRUE) ~ "Violence and Weapons",
    grepl("criminal",category,ignore.case = TRUE) ~ "Criminal Damage",
    grepl("public|anti",category,ignore.case = TRUE) ~ "Public Disorder",
    grepl("drugs",category,ignore.case = TRUE) ~ "Drugs",
    grepl("other-crime",category,ignore.case = TRUE) ~ "Other"
  ))
```

Due to the variable of weather being month a variable within crime was created for ease in merging both datasets, while making sure that all dates are in the correct format just in case there was an error somewhere. Using this variable crime totals was then formed per month along with crime type totals per month. These are then later combined into the monthly crime dataset. This dataset contains all the variables that will be used in future analysis when comparing to monthly weather data in which it is merged into to create monthly_all. The data we will use to visualise and find the relationship.

```r
crime$month <- floor_date(crime$date, "month")

crime_monthly_totals <- crime %>%
```

```
  group_by(month) %>%
  summarise(crime_total = n())

crime_monthly_type <- crime %>%
  group_by(month,Type) %>%
  summarise(type_count=n(),.groups = "drop") %>%
  group_by(month) %>%
  mutate(prop = type_count/sum(type_count)) %>%
  ungroup()


crime_monthly_outcome <- crime %>%
  group_by(month, outcome_status) %>%
  summarise(n = n(),.groups = "drop") %>%
  group_by(month) %>%
  mutate(prop = n / sum(n)) %>%
  ungroup()
```

```
crime_monthly_binded <- left_join(crime_monthly_type,crime_monthly_totals,by = "month")
```

```
monthly_all <- left_join(crime_monthly_binded,weather_monthly,by = "month")
monthly_num <- monthly_all %>%
  select(where(is.numeric))
```

Below we check all the different names of the outcome columns. Here we can see there are 11 different possibilities, keeping all 11 would create very messy and large tables therefore many of the outcomes will be reduced into 5 appropriate categories. These new outcome categories reflect the same data as the the 11 different possiblities but in a more refined way, making interpretation much easier.

```
EDA_w <- weather
EDA_c <- crime

unique(EDA_c$outcome_status)
```

```
##  [1] NA
##  [2] "Investigation complete; no suspect identified"
##  [3] "Status update unavailable"
##  [4] "Unable to prosecute suspect"
##  [5] "Court result unavailable"
##  [6] "Local resolution"
##  [7] "Formal action is not in the public interest"
##  [8] "Action to be taken by another organisation"
##  [9] "Awaiting court outcome"
## [10] "Offender given a caution"
## [11] "Further investigation is not in the public interest"
## [12] "Under investigation"
```

```
str(EDA_c$outcome_status)
```

```
##  chr [1:5956] NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA ...
```

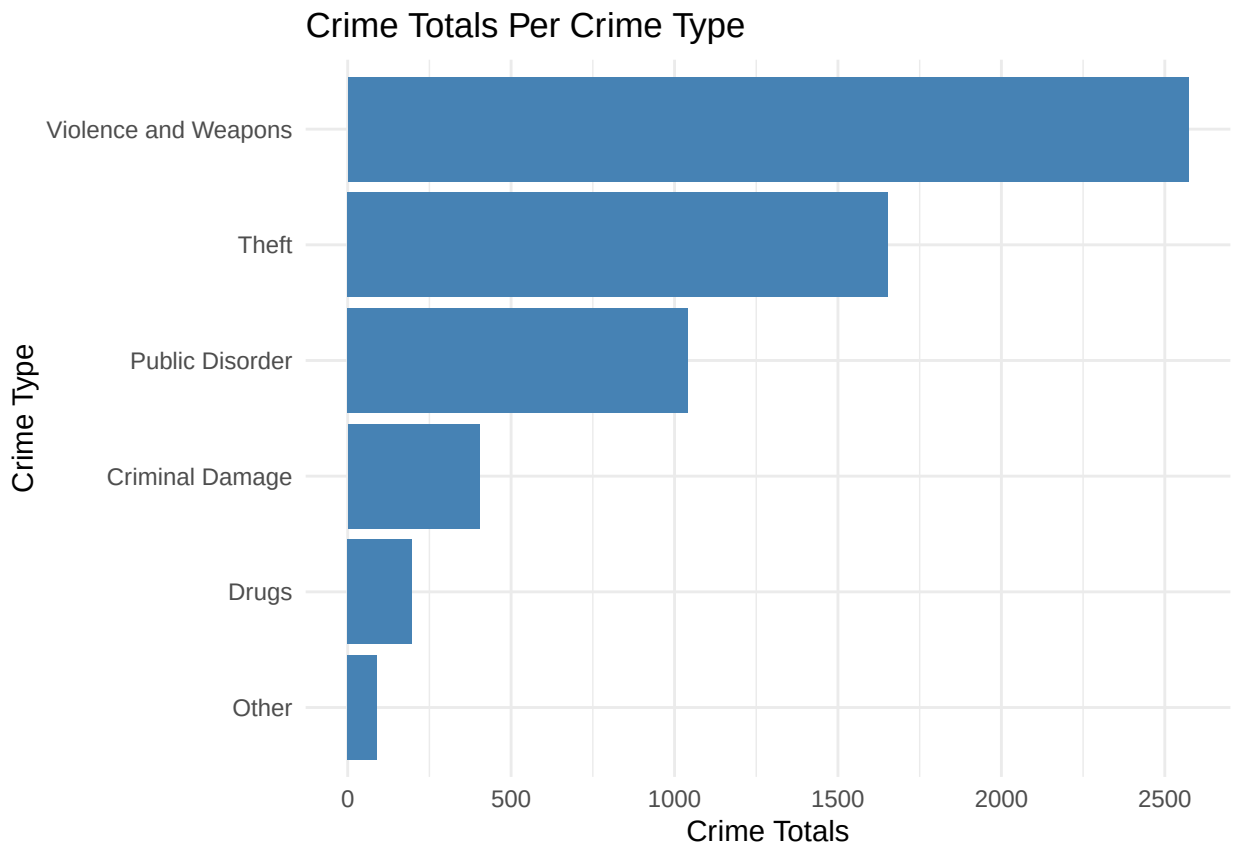
```r
EDA_c$outcome_status[is.na(EDA_c$outcome_status)] <- "Unknown"
EDA_c <- EDA_c %>%
  mutate(outcome = case_when(
    grepl("unavailable|unknown",outcome_status,ignore.case = TRUE) ~ "Unknown",
    grepl("awaiting|under",outcome_status,ignore.case = TRUE) ~ "Ongoing",
    grepl("unable|public|complete",outcome_status,ignore.case = TRUE) ~ "No Action",
    grepl("caution|resolution|organisation",outcome_status,ignore.case = TRUE) ~ "Action Taken"
  ))
unique(EDA_c$outcome)
```

```
## [1] "Unknown"      "No Action"    "Action Taken" "Ongoing"
```

##Exploratory Analysis

This section will begin by exploring the crime dataset. This dataset consists of 5956 rows and 16 columns including the changes made so far. This dataset explores crime within Colchester within 2025 including the outcomes category of crime and location of the crime. Once exploratory analysis has concluded on the crime data analysis will be conducted on the weather dataset. This dataset has the weather for all of Colchester in 2025.To start off a bar chart has been created to show the amount of crime per crime type.

```r
type_count <- EDA_c %>%
  count(Type)
ggplot(type_count,aes(x = n, y = reorder(Type, n))) +
  geom_col(fill = "steelblue") +
  scale_x_continuous(breaks = seq(0,max(type_count$n), by = 500))+
  labs(x = "Crime Totals", y = "Crime Type",title = "Crime Totals Per Crime Type ") +
  theme_minimal()
```



The bar chart shows that majority of the crimes in Colchester were to do with violence and weapons, with the least amount being due to other unnamed crimes. As can be observed there are over 750 more violent and weapons crime than the next biggest in theft. Theft itself then has quite a large margin until the next furthest with public order and then drugs just below that in 4th.

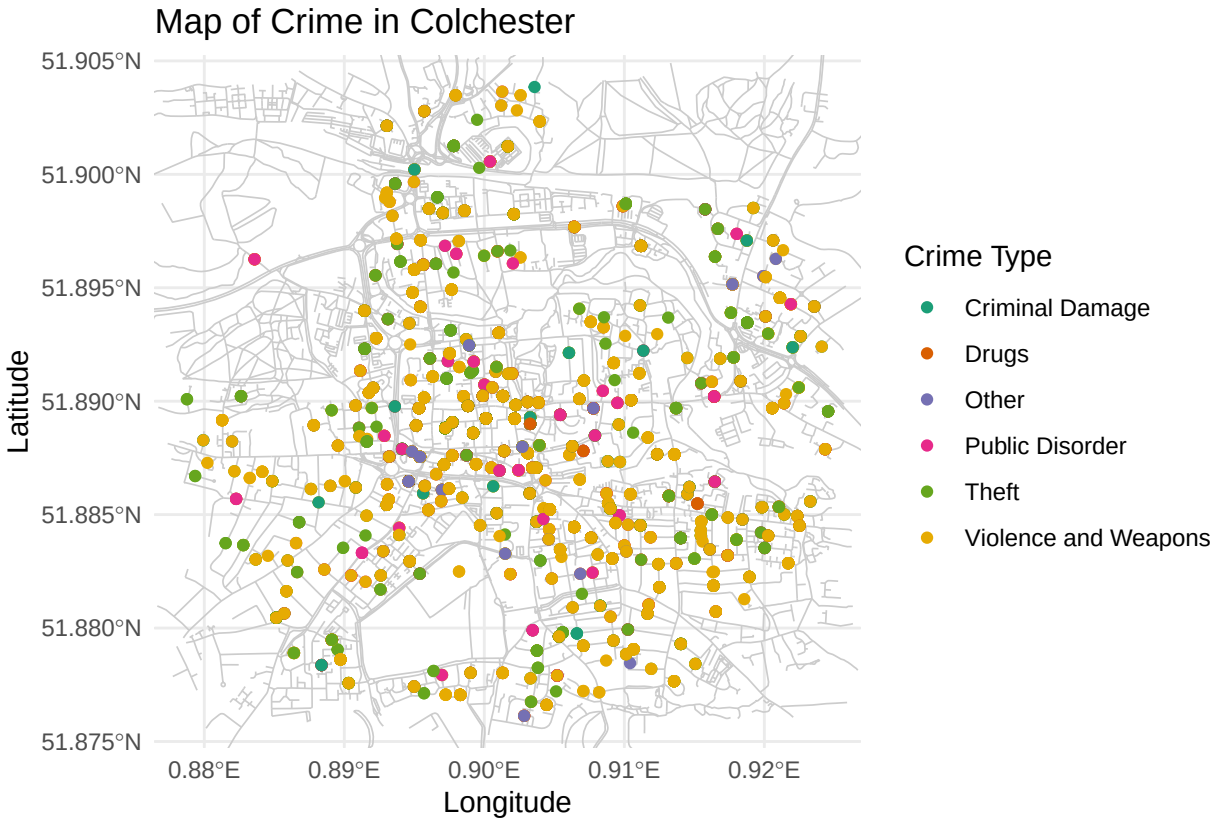
```
c_tab <- table(EDA_c$Type,EDA_c$outcome)
c_prop <- prop.table(c_tab)
c_perc <- 100*c_prop
c_perc_total <- addmargins(c_perc)
knitr::kable(round(c_perc_total, 2))
```

	Action Taken	No Action	Ongoing	Unknown	Sum
Criminal Damage	0.39	5.19	0.59	0.60	6.77
Drugs	1.66	0.62	0.71	0.32	3.31
Other	0.17	0.67	0.37	0.29	1.49
Public Disorder	0.32	5.57	0.91	10.70	17.49
Theft	0.59	22.31	2.74	2.12	27.75
Violence and Weapons	2.92	31.58	5.71	2.97	43.18
Sum	6.04	65.95	11.01	16.99	100.00

As seen by this table majority of crimes conclude without any action by a considerable margin with 65.95% of crimes ending without action. This suggests that they are minor offenses that do not have any attached punishment. This is a similar case for criminal damage and theft. Due to the relatively high number of no action its fair to assume that majority of crime is petty.

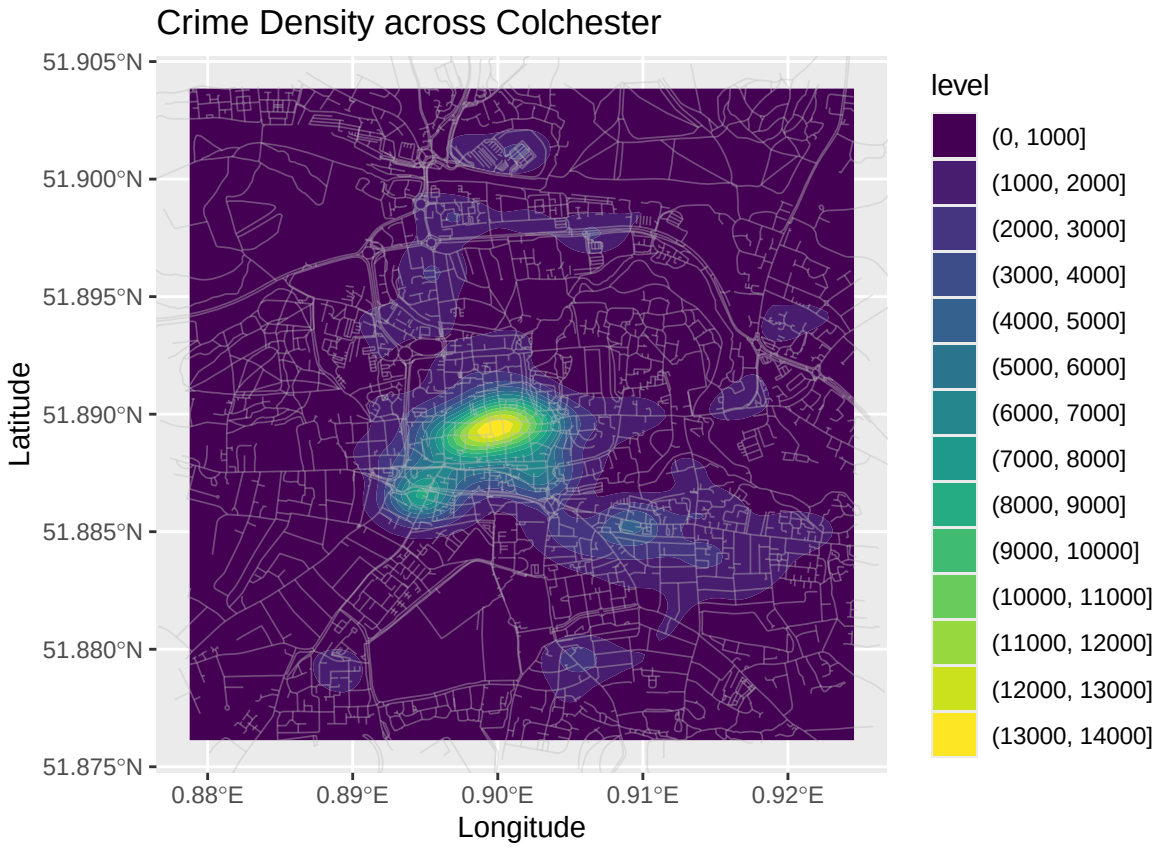
As can be seen there is a significant number of violent and weapon crimes within the city centre of Colchester. A similar pattern is recognised within public disorder, this is likely due to the number of pubs within the area. However majority of theft is found around in the suburbs which is likely due to the majority of houses being outside of the city centre and into the suburbs.

```
ggplot() +
  geom_sf(data = colchester_map$osm_lines,
          color = "grey80",
          linewidth = 0.3) +
  geom_point(
    data = EDA_c,
    aes(x = long, y = lat, colour = Type),
    alpha = 1,
    size = 1.5
  ) +
  coord_sf(xlim = c(area["xmin"],area["xmax"]),ylim = c(area["ymin"],area["ymax"])) +
  scale_colour_brewer(palette = "Dark2") +
  labs(title = "Map of Crime in Colchester", x = "Longitude", y = "Latitude",colour = "Crime Type") +
  theme_minimal()
```



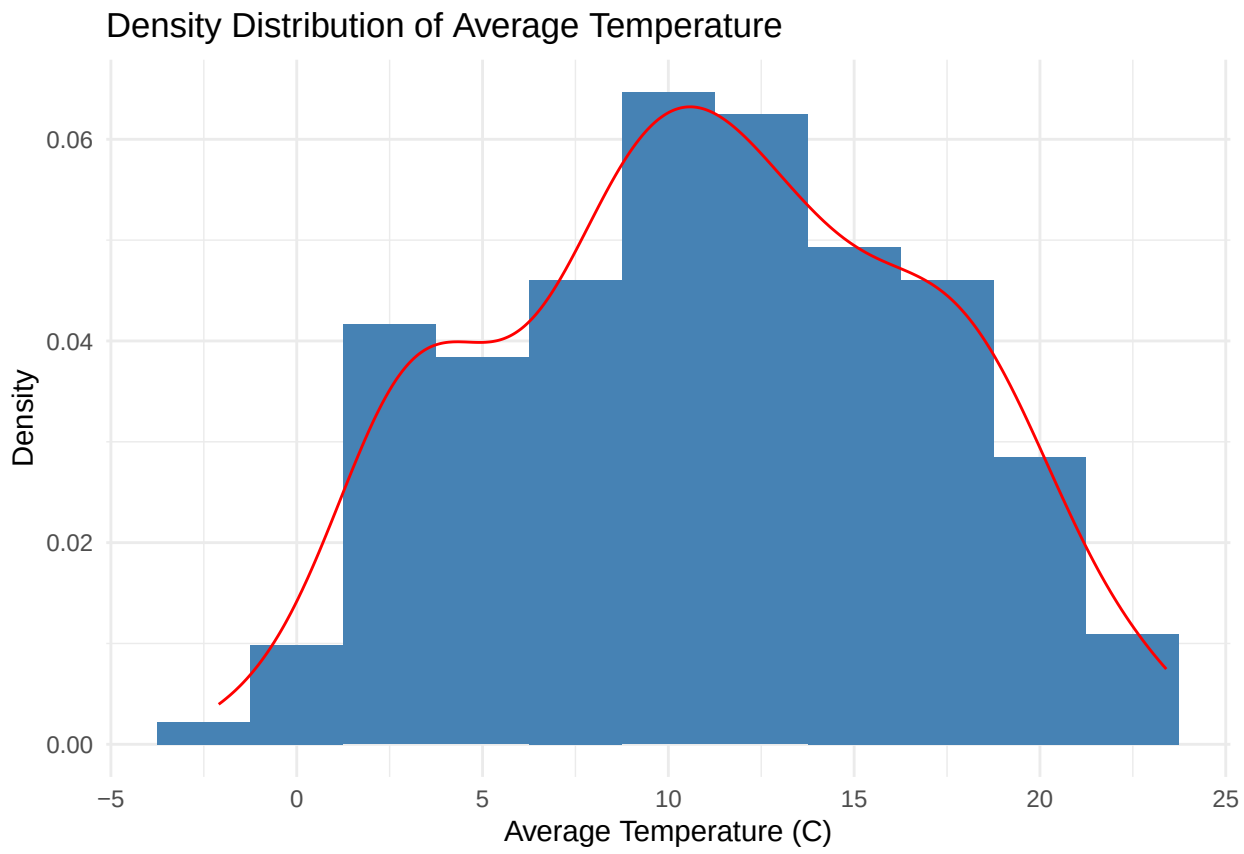
The majority of crime occurs within the city centre. This is proven by the density chart below, which shows a very high amount of crimes in the city centre area with very few spread out into the suburban areas. The density distribution is highly driven by the high proportion of violent and weapon based crimes compared to other types with majority of violent crimes being within the city centre. There are a few scatters of mildly higher density within the suburban areas these are likely due to high theft rates and violent crime and weapons existing there.

```
ggplot() +
  stat_density_2d_filled(data = EDA_c, aes(x = long, y = lat), alpha = 1,inherit.aes = FALSE,colour = NA
  geom_sf(data = colchester_map$osm_lines,colour = "grey",linewidth = 0.3,alpha = 0.4) +
  coord_sf(xlim = c(area["xmin"],area["xmax"]),ylim = c(area["ymin"],area["ymax"])) +
  labs(x = "Longitude",y = "Latitude",title = "Crime Density across Colchester")
```



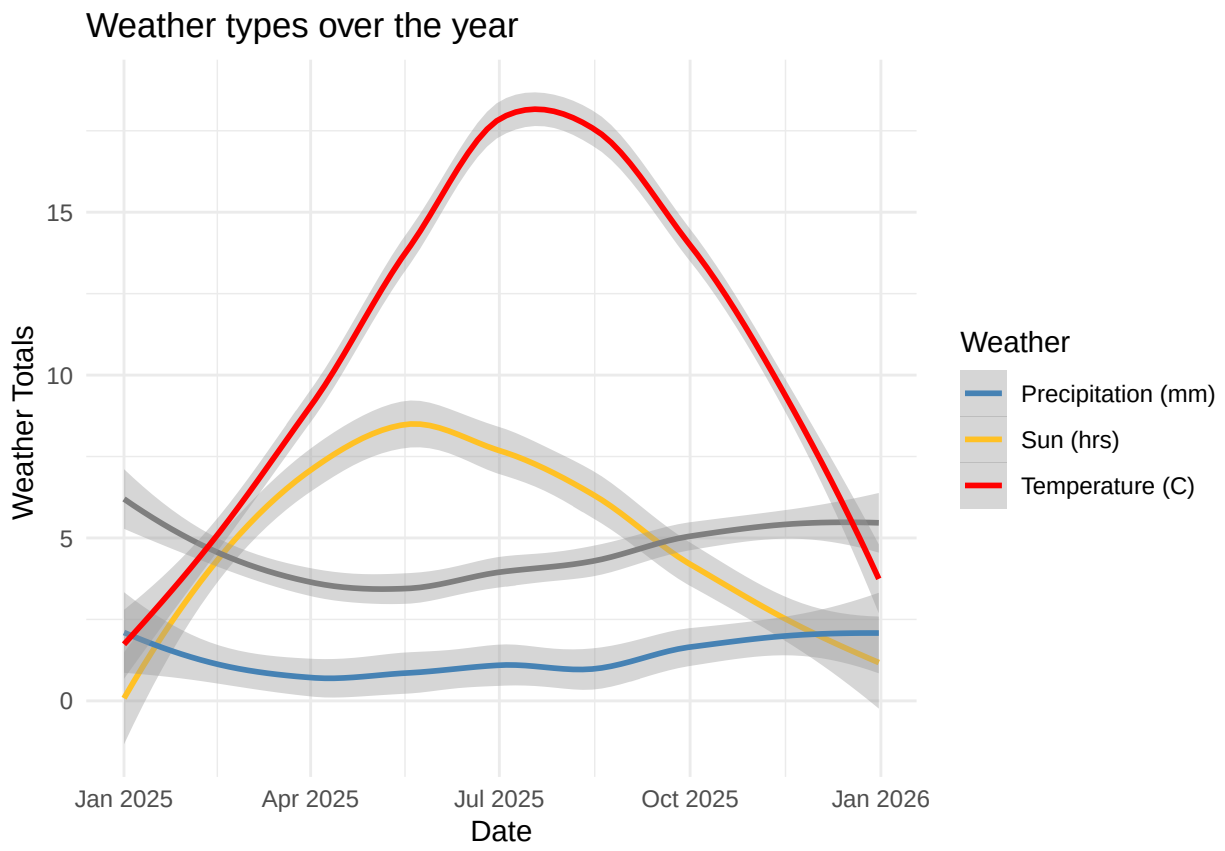
Looking at the distribution of temperatures as in the histogram below shows that it is not majorly skewed with it being a relatively normal distribution around 10 degrees. With 10 degrees being the highest density meaning majority of days the temperature average is 10 degrees. It also shows us that no day has an average temperature above 25 and none below -4 degrees.

```
ggplot(EDA_w,aes(TemperatureCAvg)) +
  geom_histogram(aes(y = after_stat(density)),binwidth = 2.5,fill = "steelblue") +
  geom_density(colour = "red") +
  labs(title = "Density Distribution of Average Temperature", x = "Average Temperature (C)", y = "Densi
  theme_minimal()
```



The smoothed distribution of weather types over the year shows many relationships between each of the 4 variables. Firstly precipitation and clouds almost map each other perfectly, which is likely due to the dependence on clouds for which precipitation to occur. There is also a slight reverse pattern of clouds and precipitation with temperature. This combined with the positive relationship between sun and temperature confirms that when its sunny there will be an increase in heat with which when its cloudy or precipitation is occurring it cannot be sunny.

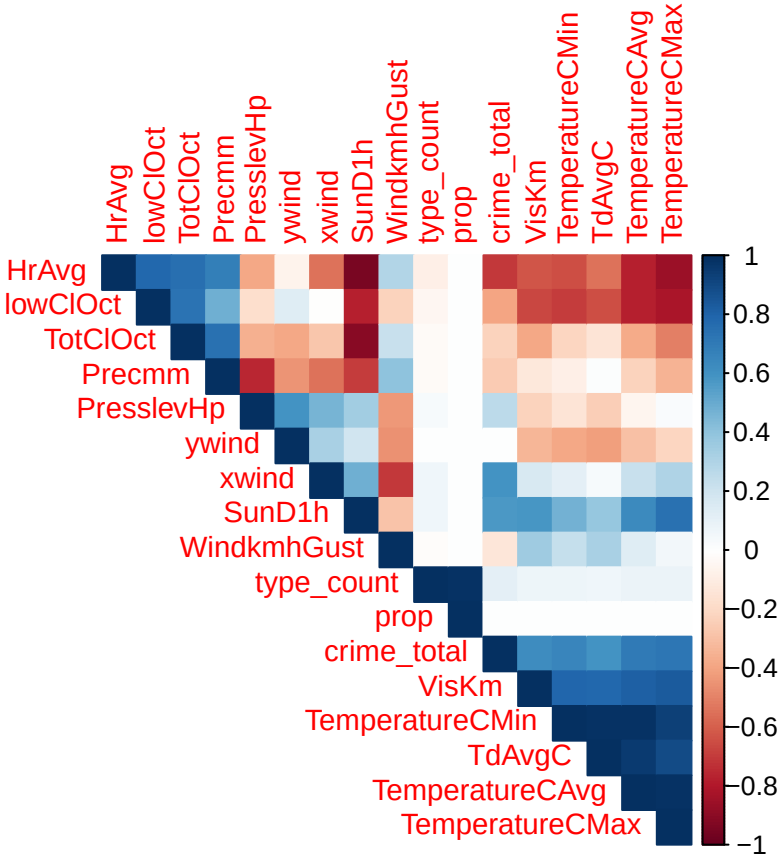
```
ggplot(EDA_w, aes(Date)) +
  geom_smooth(aes(y = SunDlh, colour = "Sun (hrs)"), na.rm = TRUE) +
  geom_smooth(aes(y = Precmm, colour = "Precipitation (mm)")) +
  geom_smooth(aes(y = TotClOct, colour = "Cloudy (Octants)")) +
  geom_smooth(aes(y = TemperatureCAvg, colour = "Temperature (C)"))+
  scale_colour_manual(values = c("Precipitation (mm)" = "steelblue", "Sun (hrs)" = "goldenrod1","Cloudy
  labs(x = "Date",y = "Weather Totals ",title = "Weather types over the year", colour = "Weather") +
  theme_minimal()
```



#Relationship Analysis

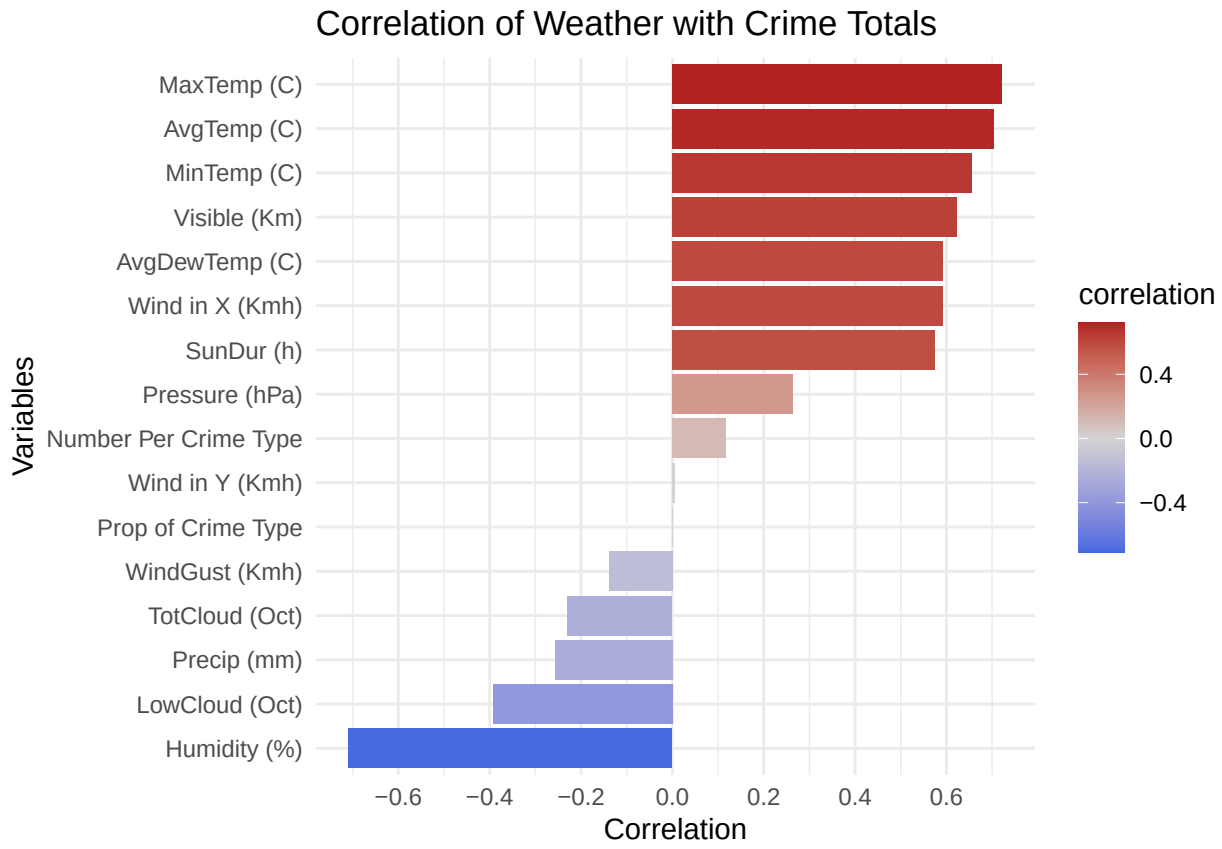
Within this section the relationship between weather and crime will be explored, including the correlations, trends and weathers relationship with the type of crime. We can see by the correlation plot below that crime type by both proportion and total have a very low correlation to any weather statistics.This informs us that crime type does not change due to different weather conditions and the proportions stay the same all year round. There are a few conditions that have a minimal effect on the number of crime per type. These minimal effects are a reflection of the change within total crime therefore showing that while they share ecological drivers the crime types are much more resistant to weather change than the crime totals.

```
corr_mat <- cor(monthly_num,use = "complete.obs")
corrplot(corr_mat,method = "color",type = "upper",tl.cex = 0.9,order = "hclust")
```



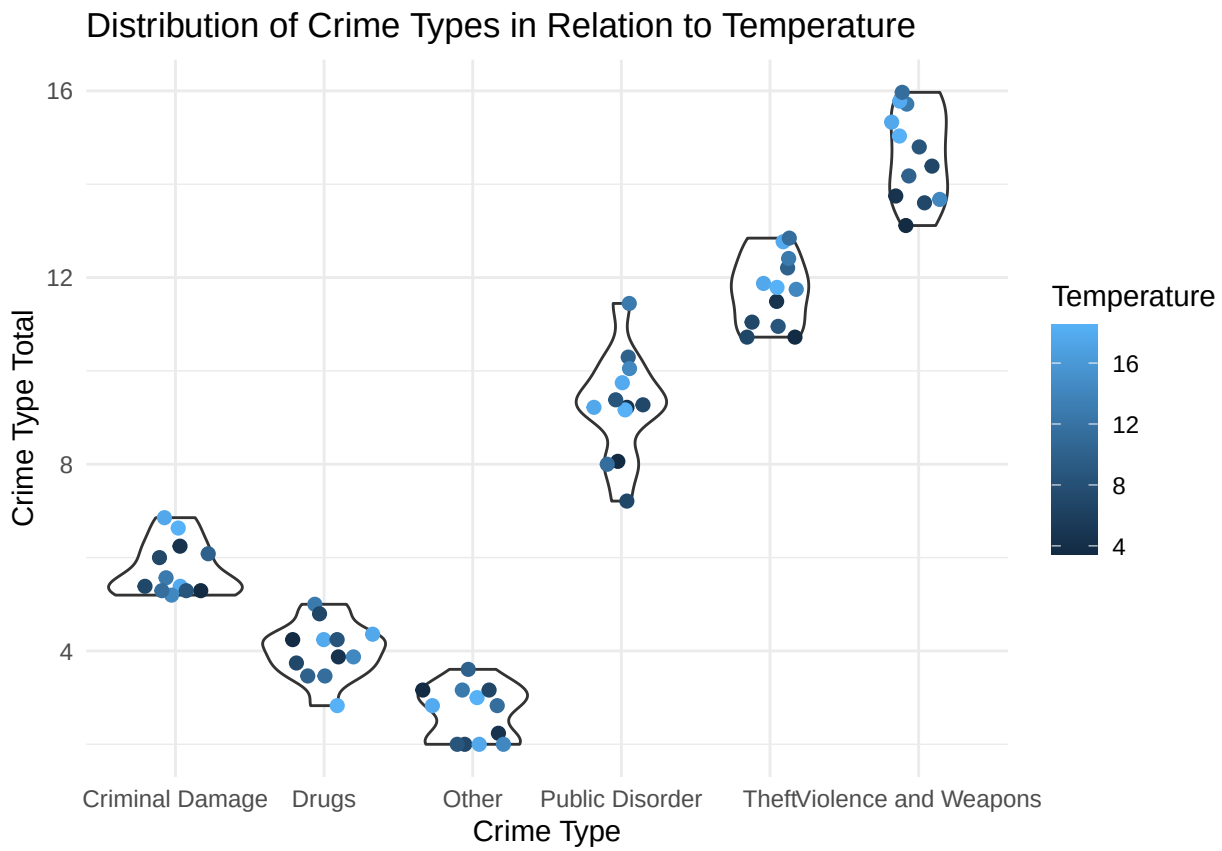
If we zoom in on just the correlations with the total amount of crime we see that maximum temperature and humidity effect crime totals the most with humidity negatively effecting crime whereas maximum temperature along with minimum and average also heavily positively correlating so that with an increase in any form of temperature there is an increase within the total amount of crime committed that month. Visibility, average temperature at dew level, wind in the x direction and sun duration also have significantly positive correlations with the crime totals therefore all helping in an increase in crime during the months they are prevalent. On the contrary however cloud coverage both low and average along with precipitation have an impact on reducing crime. As shown earlier there is an increase in cloud and precipitation within the winter months whereas sun and high temperatures are more common in the summer. Therefore due to these statistics having high impacts and correlations on crime we will conduct our relationship analysis using them.

```
pred <- monthly_num %>%
  select(-crime_total)
c_vec <- sapply(pred, function(x) cor(monthly_num$crime_total,x,use = "complete.obs"))
c_data <- data.frame(variable = names(c_vec),correlation = c_vec)
c_data <- c_data %>%
  arrange(correlation) %>%
  mutate(variable = factor(variable,levels = variable))
ggplot(c_data,aes(correlation,variable)) +
  geom_col(aes(fill = correlation)) +
  scale_y_discrete(labels = c(TemperatureCAvg = "AvgTemp (C)", TemperatureCMax = "MaxTemp (C)", Tempera
 scale_x_continuous(breaks = seq(-1, 1, by = 0.2)) +
  scale_fill_gradient2(low = "royalblue",mid = "lightgrey",high = "firebrick") +
  labs(x = "Correlation",y = "Variables",title = "Correlation of Weather with Crime Totals") +
  theme_minimal()
```



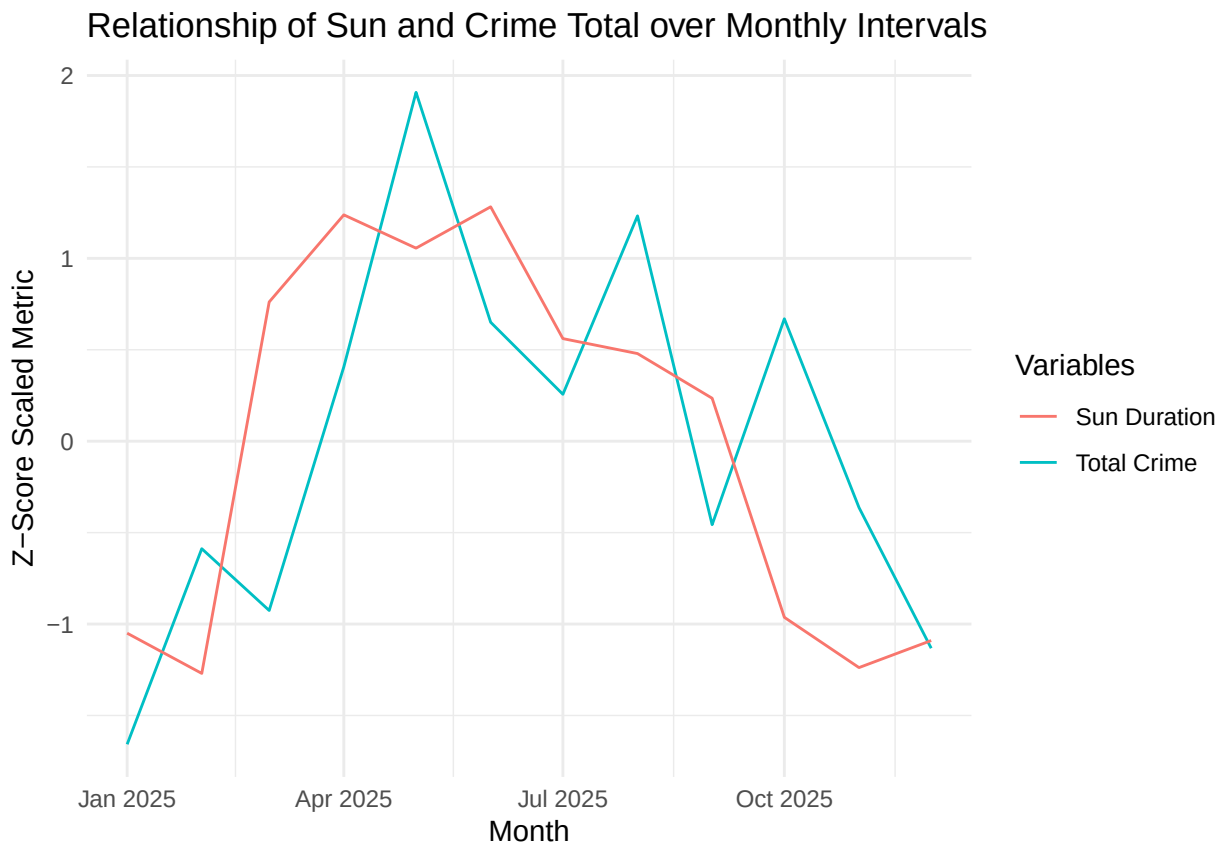
The violin plot below shows the relationship between temperature and the type of crime committed. As can be seen majority of the higher occurrence crime types, the higher average temperature months are with the higher crime counts for the category, however within the lower frequency crime types it is less correlated. With many colder months being both top and bottom along with the warmer months. The medians of drugs and other crime are left skewed while criminal damage is right skewed.

```
ggplot(monthly_all, aes(x = Type, y = sqrt(type_count))) +
  geom_violin() +
  geom_sina(alpha = 1,size = 2,aes(colour = TemperatureCAvg)) +
  labs(title = "Distribution of Crime Types in Relation to Temperature",x = "Crime Type",y = "Crime Typ
  theme_minimal()
```



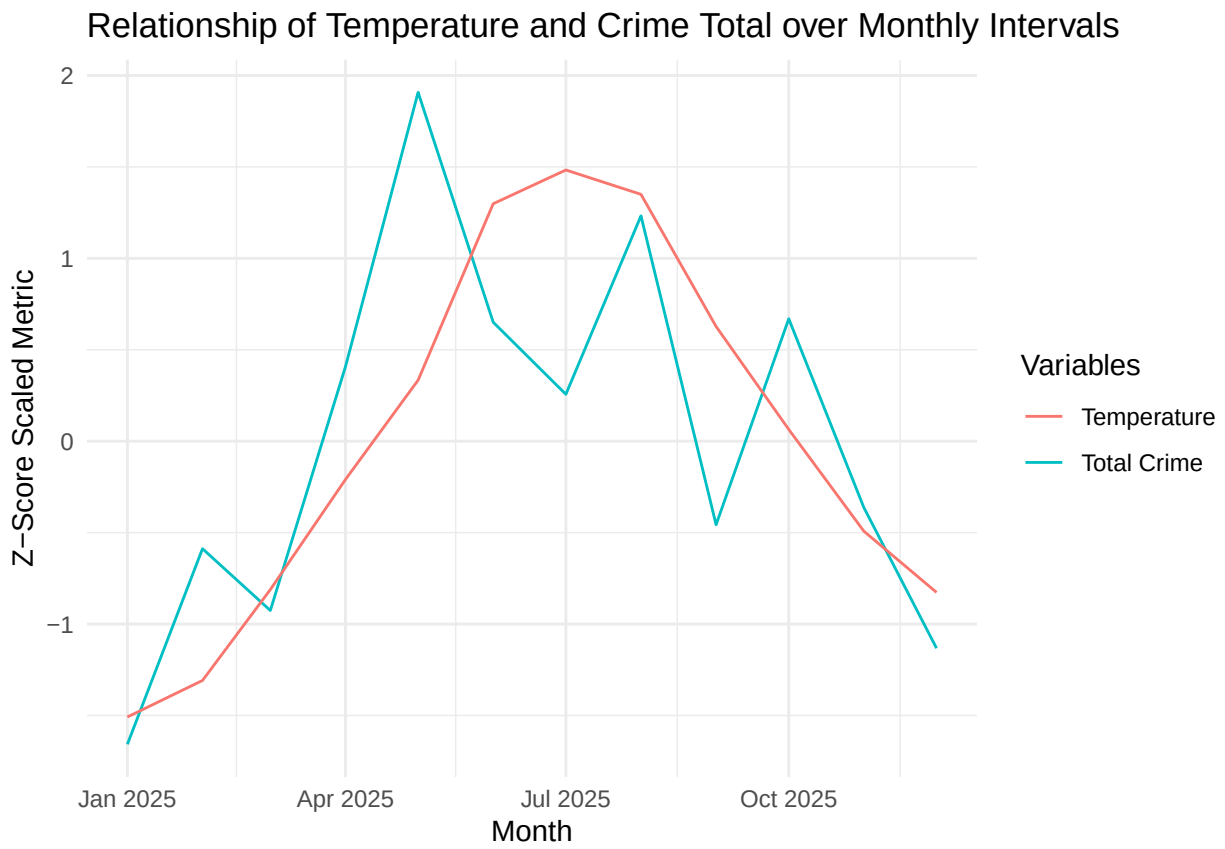
After analysing the crime type counts by temperature the time relationship between the other high correlating variables will be assessed. Firstly the sun duration shall be looked at and as we can see despite having a high correlation over the year they do not look very similar over the 12 months within the year. Sun duration models the crime relatively well although not perfectly since it tends to spike at the wrong times with a little lag from crime to catch up. This therefore informs that sun did not have a major impact on crime within Colchester in 2025 but however did effect the crime over the course of the year.

```
ggplot(monthly_all,aes(x = month)) +
  geom_line(aes(y = scale(crime_total),colour = "Total Crime")) +
  geom_line(aes(y = scale(SunD1h),colour = "Sun Duration")) +
  labs(x = "Month", y = "Z-Score Scaled Metric", title = "Relationship of Sun and Crime Total over Montl
  theme_minimal()
```



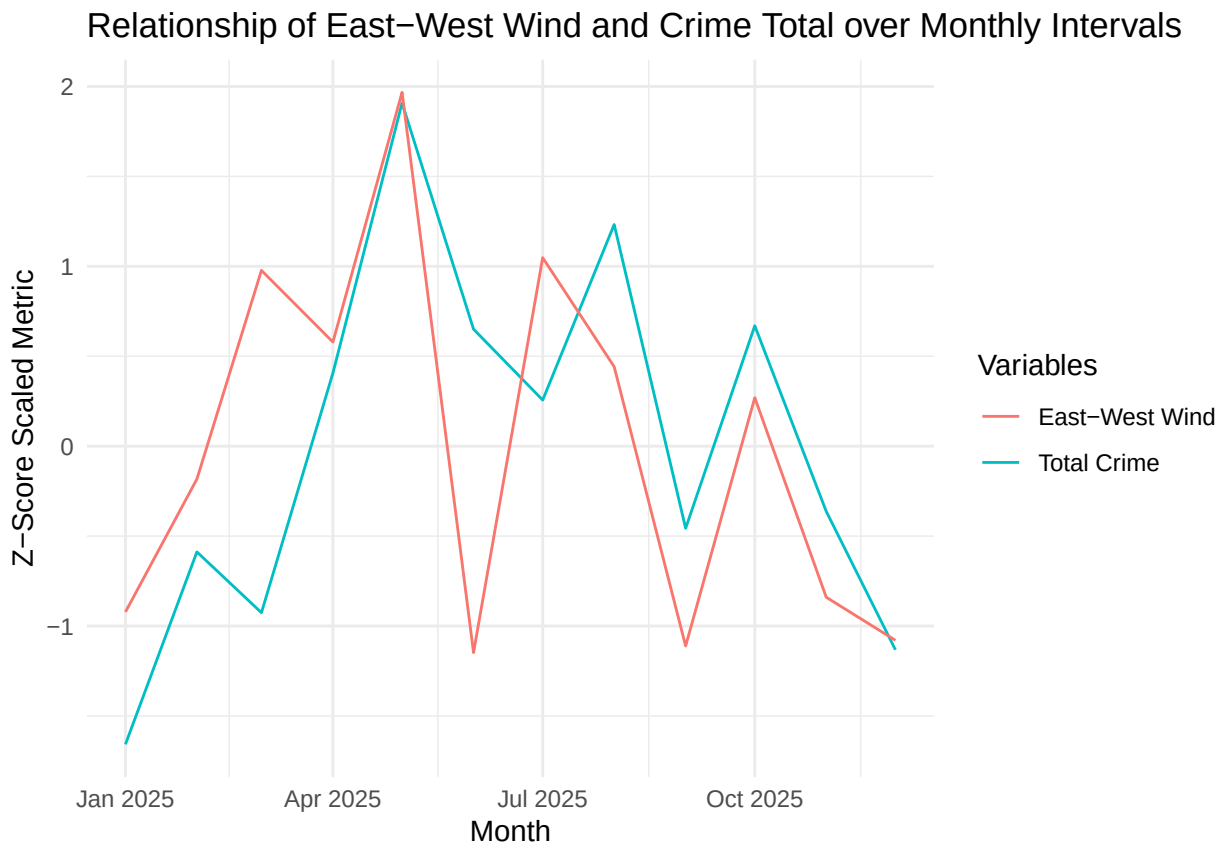
The next time series plot shows the average temperature against total crime. Maximum and minimum temperature will not be used due to their similarity with average temperature and therefore average temperature will allow us to assess the other temperature variables without separate plots. As can be seen by the graph temperature does not map at all well towards total crime with the temperatures increasing until July and then decreasing for the remainder of the year while crime peaks in may and spikily declines since. This could be due to at a certain point the weather becomes too hot influencing the crime to then decrease which would explain the largest and also final spike within the total crime line as these occur at similar temperatures.

```
ggplot(monthly_all,aes(x = month)) +
  geom_line(aes(y = scale(crime_total),colour = "Total Crime")) +
  geom_line(aes(y = scale(TemperatureCAvg),colour = "Temperature")) +
  labs(x = "Month", y = "Z-Score Scaled Metric",title = "Relationship of Temperature and Crime Total ov
  theme_minimal()
```



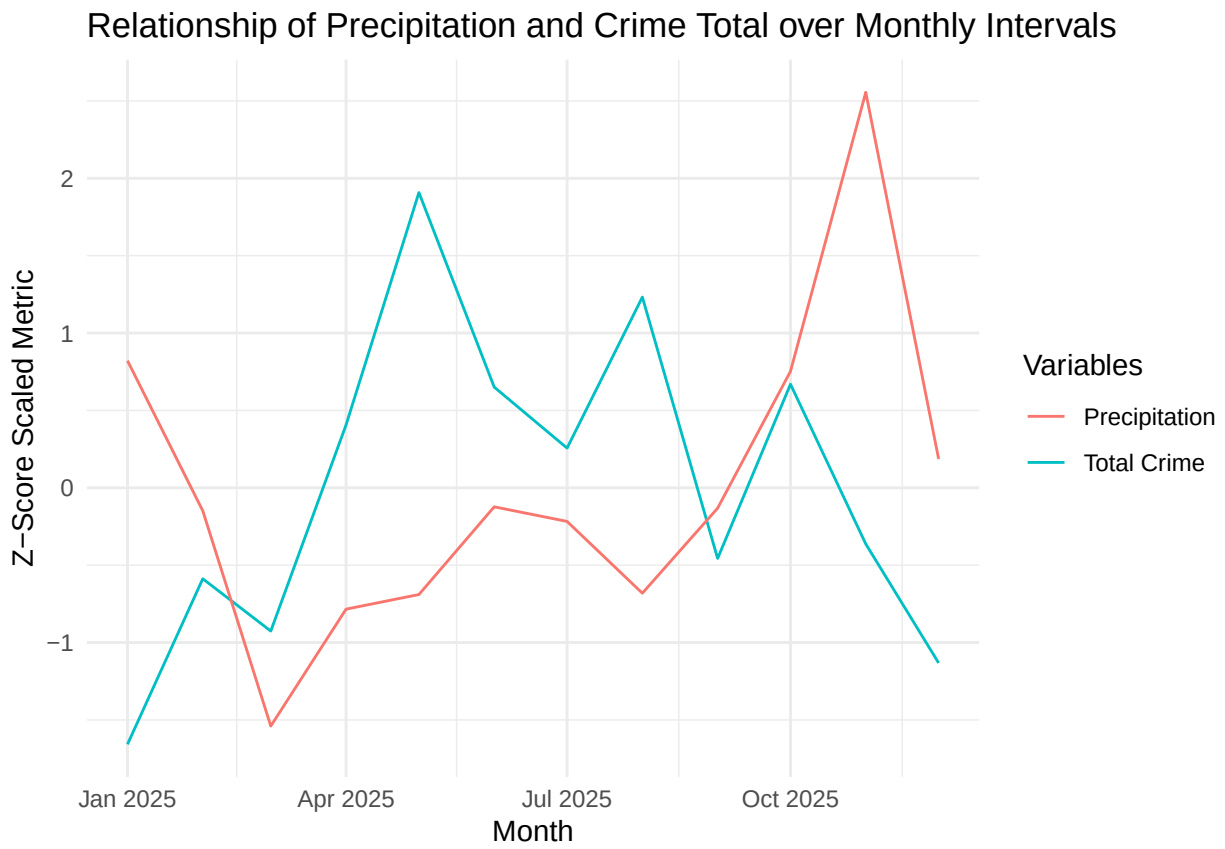
Another variable that contained a significant correlation towards the total number of crime was wind within the x direction, otherwise interpreted as East-West. With the time series plot below it concludes that wind has a major correlation and impact on the total number of crime with peaks and drops showing a near-perfect match. The biggest difference is during the month of July where there is a much lower amount of wind whereas crime total does not majorly fall. The peaks of wind in may and October are perfect or near perfect to the peaks within the total crime presenting the impact of which East-West wind has upon the number of crimes committed.

```
ggplot(monthly_all,aes(x = month)) +
  geom_line(aes(y = scale(crime_total),colour = "Total Crime")) +
  geom_line(aes(y = scale(xwind),colour = "East-West Wind")) +
  labs(x = "Month", y = "Z-Score Scaled Metric",title = "Relationship of East-West Wind and Crime Total
  theme_minimal()
```



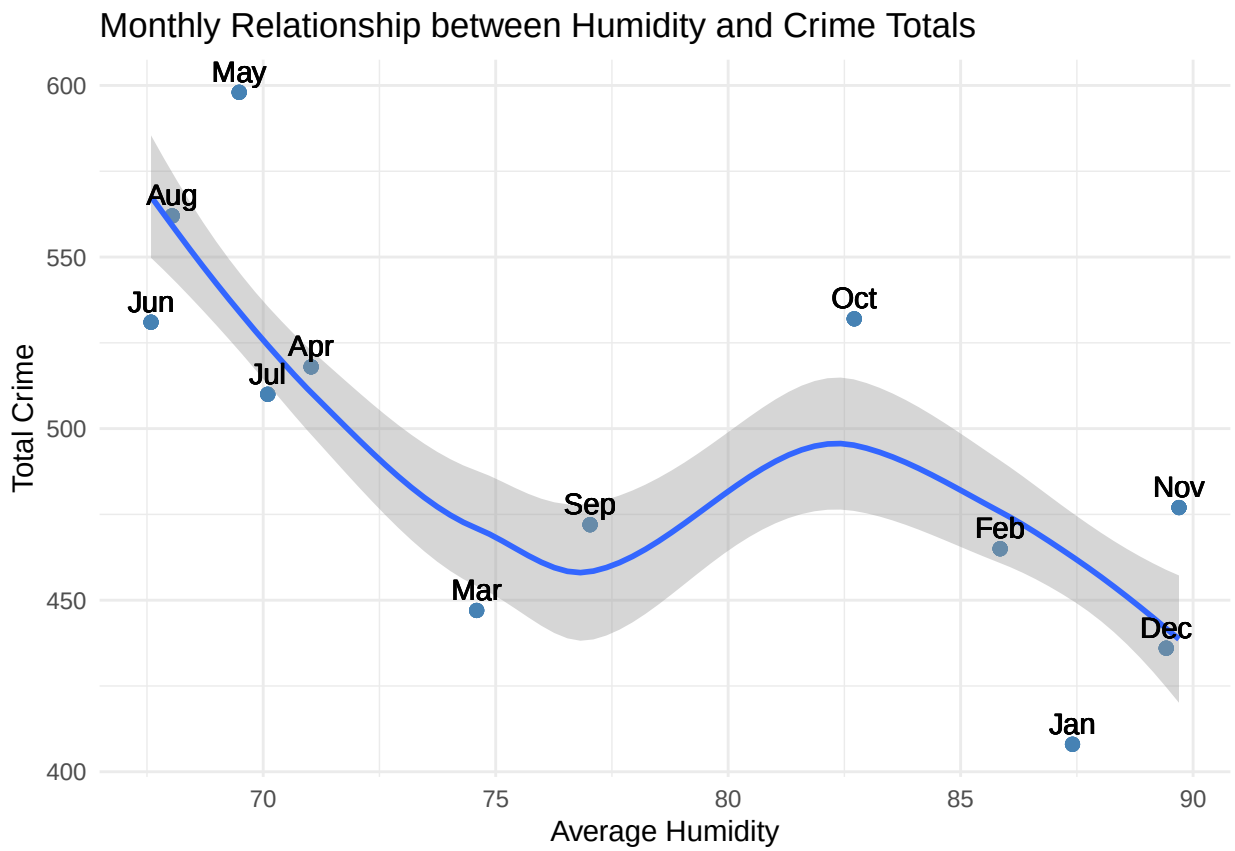
Precipitation was also a variable with which had a significant correlation with crime number however in the negative direction. As shown by the time series plot precipitation does not seem to have much impact on the total crime levels either, this is shown by on several occasions mimicking spikes within the crime total and not showing the opposite each month. Instead of an upward spike of total crime with a downward spike of precipitation, or the contrary, in some instances precipitation matches the patterns within total crime and at other points does spike in the opposite direction.

```
ggplot(monthly_all,aes(x = month)) +
  geom_line(aes(y = scale(crime_total),colour = "Total Crime")) +
  geom_line(aes(y = scale(Precmm),colour = "Precipitation")) +
  labs(x = "Month", y = "Z-Score Scaled Metric",title = "Relationship of Precipitation and Crime Total
  theme_minimal()
```



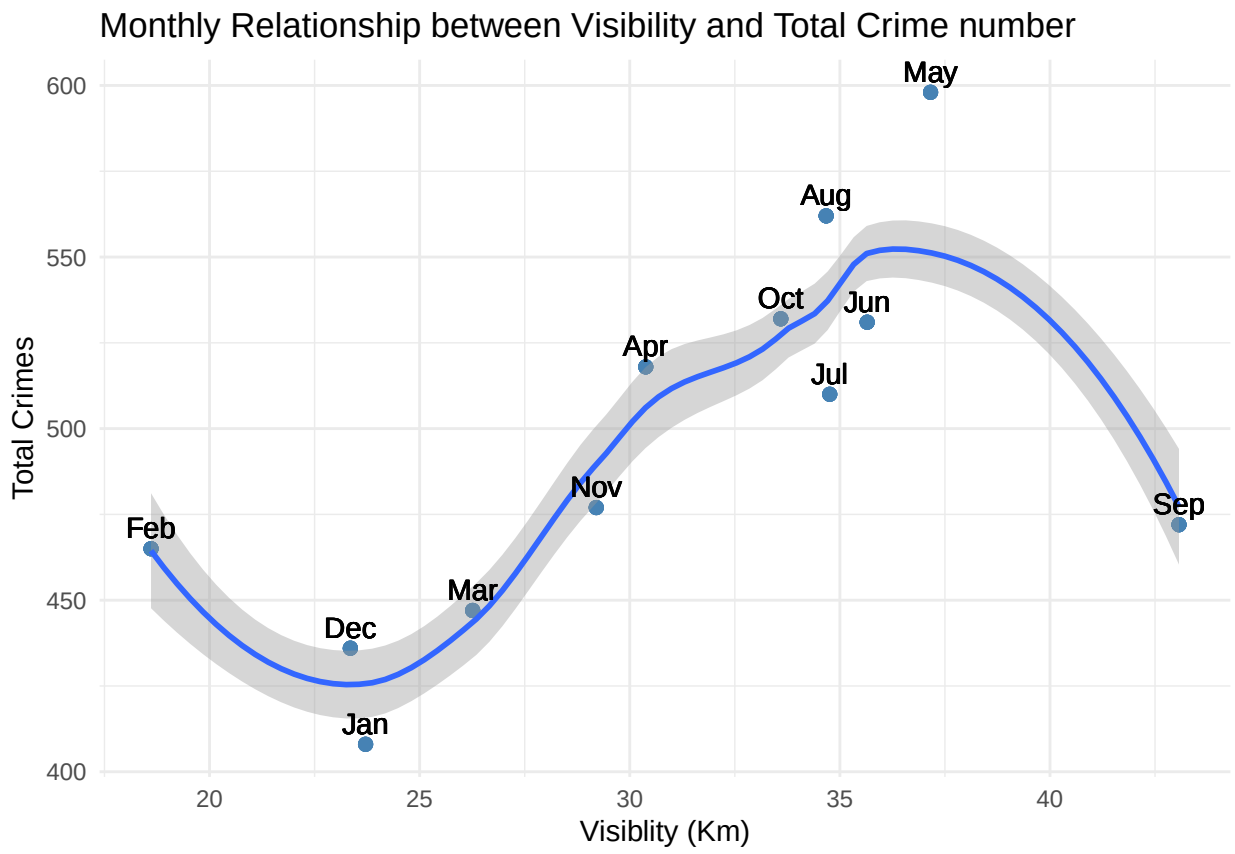
Next we will look at scatter plots with smoothed lines to determine whether there is a monthly relationship between humidity and total crime. Examining the graph below we can see that there is a high crime count within the hotter summer months where there is much lower humidity whereas within the colder months where there is much more humidity there is far less crimes. There is a little spike in crimes around the 82.5% humidity level due to a higher crime rate within october caused by other factors. This suggests that while humidity does have a big impact on crime totals it is not the only factor that effects it.

```
ggplot(monthly_all,aes(HrAvg,crime_total)) +
  geom_point(colour = "steelblue",size = 2) +
  geom_smooth() +
  geom_text(aes(label = format(month,"%b")),hjust = 0.5,vjust = -0.5) +
  labs(title = "Monthly Relationship between Humidity and Crime Totals", y ="Total Crime", x = "Average
  theme_minimal()
```



The next smoothed scatter plot defines the relationship between visibility and crime total. This plot is very different from the previous as it shows low visibility comes with low crime rates and is often within the the colder winter months where there is more turbulent weather such as precipitation. From 25km to 35km there is an increase and positive trend between crime total and visibility although this then tails off to support the data collected in September. This shows that while for lower visibility it has a impactful relationship with crime total once a certain visibility is reached this then no longer is a meaningful factor.

```
ggplot(monthly_all,aes(VisKm,crime_total)) +
  geom_point(colour = "steelblue",size = 2) +
  geom_smooth() +
  geom_text(aes(label = format(month,"%b")),hjust = 0.5,vjust = -0.5) +
  labs(title = "Monthly Relationship between Visibility and Total Crime number", x = "Visiblity (Km)", y
  theme_minimal()
```



#Conclusion and Interpretation

The data visualisation conducted above has showed that while many factors contain strong correlations with total crime only few are actually impacting the rates of crime occurring within Colchester in 2025. The weather variables with the biggest impact on crime are humidity, visibility and sunshine.It was found that an increase in sunshine and visibility generally means more crime except in an extreme case for visibility. Whereas the higher the humidity the lower the crime rate since a higher humidity is linked to the winter months which generally had less crime than the hotter months.

Spring was found as the most dangerous month likely due to its reasonable temperature and social effects of people effectively having to hibernate within the winter months and wanting to go out again. Therefore where there is more opportunity with more people out there is more chance for crime to occur and with more chance there is more crime occurring in general.

It was also found that for the crime types, the 3 crime types with the highest frequency all followed a pattern of lower counts being within the colder months while higher counts were within the hotter months. This was not the case however for the lower frequency crime types where there was no distinct pattern between more crime and average temperature.

Overall it can be concluded that there is a relationship between weather and crime within Colchester in 2025. With conditions often found within the hot months of Summer increasing the total of crimes occurring while other weather conditions such as humidity which is often found within the winter months correlates with less crime. However the seasons with the most crime are found in-between both when East-West wind is at its peak within Spring.

Therefore Colchester is more dangerous within the Summer month than within the colder months however it is not the most dangerous with Spring containing the highest number of crimes, with a high number being recorded in May.